

## THE DEMAND SPACE OF PLANAR TRUSS CONNECTIONS

### 4.1 INTRODUCTION

Understanding the distribution of structural demand for truss elements is important for controlling and understanding the effect of global form and aggregated performance metrics on the complexity of its realization. Consider the truss structure in Fig. 4.1, a simplified version of the timber scaffolding for the Salginatobel bridge, constructed by Richard Coray. The internal forces are heavily dominated by two arching systems in the top and bottom “chords” of the system, highlighted by the line intensity in the axial force diagram. Many of the interior elements act as near zero-force members that provide lateral stability but do not contribute significantly to the primary load path; these conditions are reflected by the spread of points in the element Demand Space. Due to the unconventional geometry of the structure, a large spread of required lengths are observed on the  $y$ -axis, and the zero-force members are shown by the vertical cluster of points near the origin. This Demand Space representation can help inform the builder on the range of lengths and cross-sections necessary to construct the structure, and can provide the basis for rationalization techniques such as grouped section procurement and stock cutting to minimize cut-off waste and construction complexity.

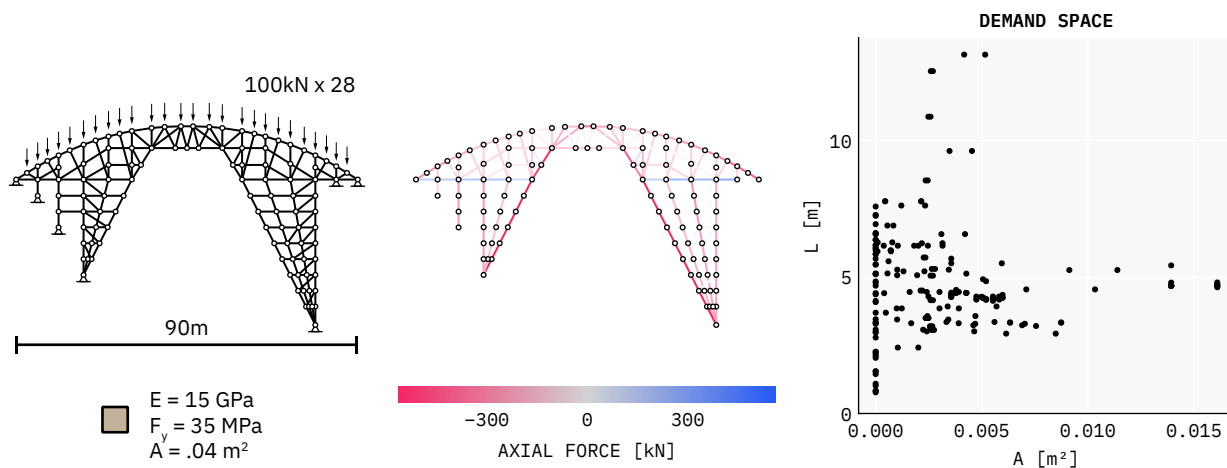


FIGURE 4.1: Truss model of the scaffolding for the Salginatobel bridge under load, and the elemental Demand Space.

The connections between axially-loaded truss elements play an equally important role in the realization of a given design. Varying magnitudes and orientations of forces must be carefully aligned and transferred between elements to provide a functioning load path from external loads to supports. Alignment of idealized force vectors to a common reference point ensures overall equilibrium without adding induced moments and stresses into the system. As the geometries of truss structures become more free-form in search of global

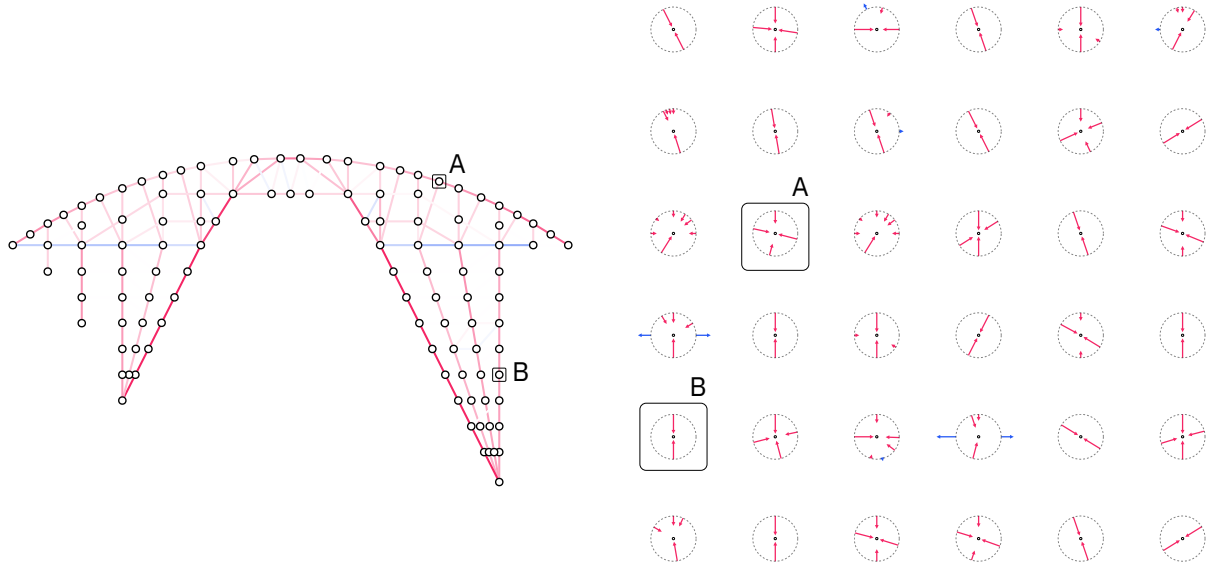


FIGURE 4.2: Forces acting at truss nodes.

material efficiency, the design of connections also play a more important role during the assembly process for alignment and member registration. For a complete understanding of the anticipated complexity of truss structures, the Demand Space of truss connections and their induced demands is required. Like that of their connected elements, this Demand Space requires a geometric representation of each node that captures all relevant information necessary for their construction, as well as an ability to measure similarity between any two objects in this space.

In Fig. 4.2, a collection of these demands acting at a subset of the truss nodes is shown, with lengths of each force arrow corresponding to relative magnitude with respect to other nodes acting on the same node. Unlike truss elements, with a clear set of requirements (area and length) that compactly describe their needs for construction, the demands of truss nodes are more difficult to capture. Not only are force magnitudes required, but their relative orientations must also be captured to understand the geometric configuration of load transfer. Further, as the valence, or the number of elements connected to a node, varies across the structure, the quantity of information required to describe the demands at a node is not fixed. A final challenge unique to truss connections is its orientation in space. A truss element with a given length under a compressive load can be described in the Demand Space by a single point, regardless of whether it is a vertical column or an inclined member in a spanning system. A proper descriptor for nodal force demands, which are defined in part by the *relative* orientations of its connected members, should also provide this invariance to rigid rotations in space.

This chapter presents a method for the characterization of nodal force demands that overcomes the challenges of dimensionality and invariance, enabling a systematic generation of the Demand Space for truss connections. The method is centered around a functional interpretation of nodal forces, through which harmonic decomposition can be used to

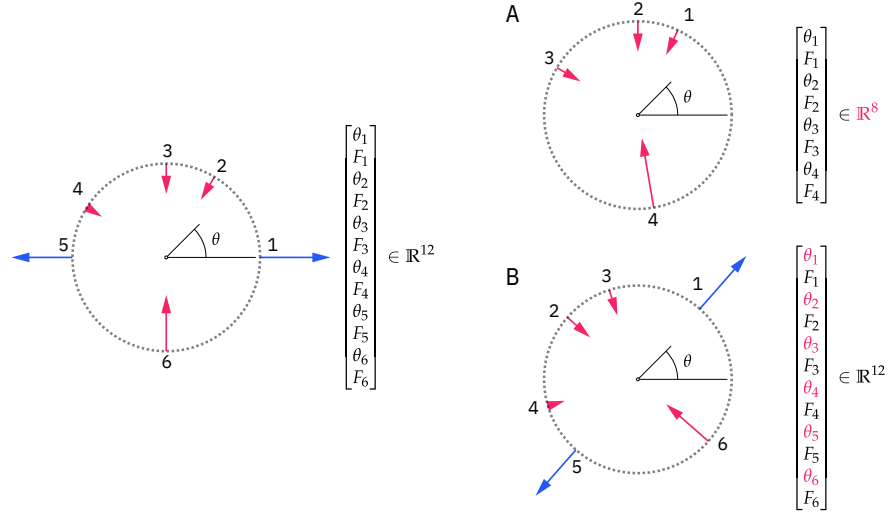


FIGURE 4.3: Directly comparing nodal demands is difficult due to both varying valence and the lack of invariance of force ordering and rotational orientation.

generate compact feature vectors that represent arbitrary node demands as points in high-dimensional space.

#### 4.2 DIMENSIONALITY AND INVARIANCE

As opposed to the Demand Space of structural elements, which are defined by a consistent and finite space of requirements (such as length, stress, and buckling), the Demand Space of truss nodes is more difficult to characterize. One primary challenge is in consistent *dimensionality*: Although the individual forces that act on a node can be clearly defined by an orientation and magnitude, the overall number of forces acting at a node vary depending on the topology of overall structure, and thus the number of independent considerations that make up the Demand Space of a truss node is not fixed. This consistent dimensionality is critical when operating within the Demand Space, where measures of complexity are captured by metrics of distance, an ill-defined concept across uneven dimensions. The other challenge lies in the lack of *invariance* in the typical ways of representing nodal force demands. Unlike the Demand Space of truss elements, which consists of fixed axes of information that are independent of the spatial position and orientation of a given element, nodal force demands must also encode the relative positions of the individual forces acting at a node. When force orientations at a node are measured through an absolute reference frame, such as a Cartesian coordinate system, a node will have a different representation after a rigid rotation in space.

For example, consider the node of valence six under load in Fig. 4.3. A naive encoding of all relevant demand information would be a concatenated vector of six magnitudes and six orientations (here, defined as angles,  $\theta$  from a common reference axis), resulting in a representative vector in  $\mathbb{R}^{12}$ . When representing a single node, this is a sufficient and compact encoding of all information relevant to the structural designer. However, if

comparison is desired with the node A in Fig. 4.3 with valence four, a distance metric must be defined that can be computed across two different dimensional spaces,  $\mathbb{R}^{12}$  and  $\mathbb{R}^8$ . Although not impossible, this would rely on computationally expensive subroutines that perform best-fit analysis between the two spaces, e.g., augmenting node A with additional zeros to match the dimensionality of the initial node, and optimizing for the location of these padding zeros. Even if this is performed, the second challenge of invariance remains. As shown by node B, which is simply a rigid counter-clockwise rotation of the original node, the entries of the feature vector representing force orientation have changed, as the characterization scheme relies on a fixed reference axis. Although a quick measure of similarity could be performed, for example, through the Euclidean distance between the two vectors, it would result in a non-zero distance for an otherwise identical force condition. A useful characterization of nodal force demands must overcome both of these barriers to find practical use in structural design and Demand Space Analysis.

#### 4.3 FUNCTIONAL REPRESENTATION OF NODAL FORCE DEMAND

We first address the challenge of dimensionality by representing nodal force demands not as compact lists of individual forces and orientations, but as the output of a smooth periodic function around the unit circle centered on the node. Given a force exerted by an element  $F_i$  connected at a node with some angle  $\phi_i$ , a smooth periodic function,  $f_i(\theta) : \theta \in [0, 2\pi) \rightarrow \mathbb{R}$ , based on a Gaussian bump can be represented by Eq. (4.1):

$$f_i(\theta) = F_i e^{-\delta(\theta - \phi_i)^2} \quad (4.1)$$

where  $\delta$  is a calibrated parameter akin to the variance of a Gaussian distribution, which represents the spread of an individual force acting at  $\phi_i$ . The value of this function is zero everywhere except near  $\phi_i$ , with a peak value of  $f_i(\phi_i) = F_i$ , and the overall nodal force demand represented by the sum of individual Gaussian bumps:

$$f(\theta) = \sum_{i=1}^n f_i(\theta) \quad (4.2)$$

Numerically, a given nodal force demand is represented by taking samples of Eq. (4.2) at a fixed number of intervals from  $0 \rightarrow 2\pi$ , giving a discrete vector-valued representation of its output,  $f \in \mathbb{R}^N$ , where  $N$  is the number of samples taken. By fixing  $N$  for a given structural model, these sampled force functions can then be taken for all nodes in the structure. An example of the functional representation of nodal demand is shown in Fig. 4.4.

As mentioned, the parameter  $\delta$  controls the degree of spread of each individual force bump, and should be calibrated with respect to the typical topological configurations acting at the nodes of the structure in question. Smaller values of  $\delta$  correspond to sharp peaks with little spread, converging to a Dirac delta function at its limit, while larger values of  $\delta$  provide a smoother transition towards each peak and a larger effect on values of  $\theta \approx \phi_i$ . An example of this effect is shown in Fig. 4.5. Smaller values of  $\delta$  allow for proper delineation between forces that act close together at the cost of smoothness at low sample resolutions,  $N$ ; larger values of  $\delta$  provide a smoother representation of individual forces, but result in more overlap between adjacent forces and a loss of clarity. The latter is not necessarily

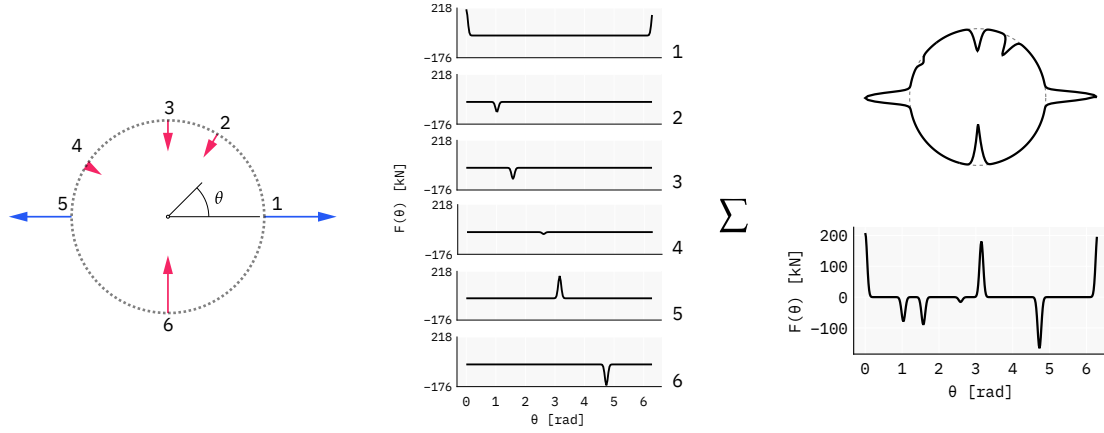


FIGURE 4.4: The functional representation of nodal force demand is a sum of individual Gaussian bumps around the unit circle, with magnitudes and orientations corresponding to each force acting at the node.

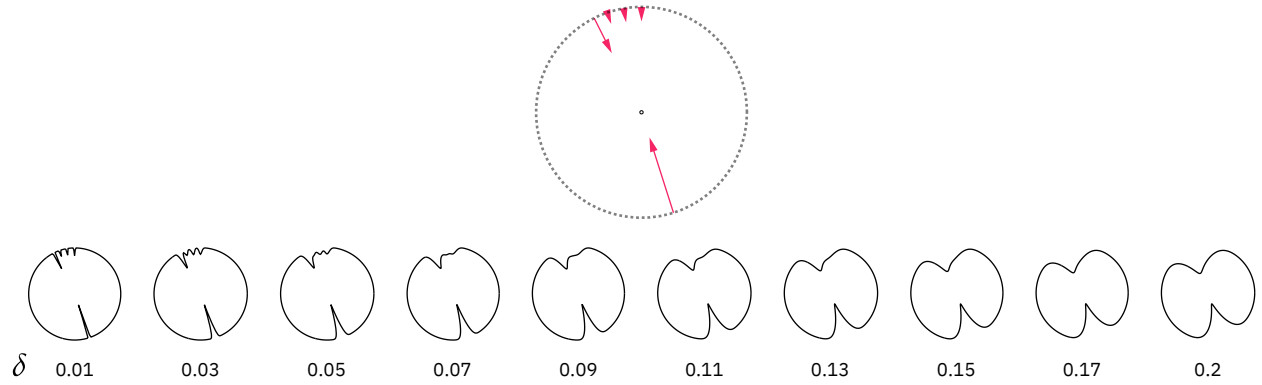


FIGURE 4.5: Effect of varying  $\delta$  on the nodal force function.

undesirable, as it naturally provides weighing of important force contributions, both in magnitude and orientation, providing a convolution effect for complex nodes.

#### 4.4 FOURIER ANALYSIS OF FORCE FUNCTIONS

The periodic function representation of nodal force demands solves the challenge of dimensionality, but still suffers from issues of rotational invariance. Two nodes with equivalent force conditions, but different orientations, will register as different curves in its functional representation, akin to a circular shift in the independent axis,  $\theta$ . Through the use of the Fourier transform, we show how rotational invariance is achieved by decomposing each force function into a set of weights for basis functions that are independent of the initial orientation of the node in question. These weights provide the compact and invariant representation needed to evaluate the similarity and uniqueness of truss connections under load.

The Fourier transform is a change in basis of an arbitrary signal or function into a weighed sum of periodic functions, namely sine and cosine functions of increasing frequencies. Developed first by Joseph Fourier in the 1800s, originally for the study of heat conduction, it has seen continued and extensive use in the fields of signal processing, differential equations, and image analysis [133–135]. It has also found applications in shape characterization for the purpose of shape recognition and similarity analysis [136–138], from which we draw inspiration for its use. In this chapter, we apply the same principles of shape characterization to that of nodal force demands in truss structures. We show how challenges that arise in generic Fourier descriptors for arbitrary shapes can be bypassed by directly representing nodal force demands as smooth functions on the unit circle, and how the results of our method can be used in practice for rationalization and anomaly detection in truss design.

This method of decomposing arbitrary information is akin to the use of basis vectors in linear algebra, where the linear independence (orthogonality) of a finite set of vectors can be used to represent *any* vector within the same space. An arbitrary point in space,  $\mathbf{p} = [x, y, z]$ , is an implicit linear combination of the unit orthogonal vectors,  $[1, 0, 0]$ ,  $[0, 1, 0]$ , and  $[0, 0, 1]$ , such that:

$$\mathbf{p} = x \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + y \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} + z \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

For an  $N$ -dimensional point, this can be extended to a weighed sum of  $N$  orthogonal basis vectors,  $\mathbf{e}_i$ , such that:

$$\mathbf{p} = \sum_{i=1}^N x_i \mathbf{e}_i$$

The Fourier transform extends this concept to the infinite-dimensional vector space of functions, where instead of unit vectors corresponding to each spatial dimension, the orthogonal bases are the set of periodic cosine functions such that:

$$f(\theta) = -A_0 + \sum_{k=1}^{\infty} A_k \cos(k\theta + a_k) \quad (4.3)$$

Here, the pair  $A_k, a_k$  represent the amplitude and phase shift of the corresponding frequency multiple,  $k$ . In its general form, this pair of numbers, represented as a complex-valued coefficient,  $X_i = A_i + ia_i$ , is the output of the Fourier transform. Since the frequencies of the cosine functions increase with  $k$ , there is a general correspondence between the features of the function under analysis and the values of the coefficients. Lower values of  $k$  correspond to broader, smooth features of the overall function, such as global valleys and peaks, while higher values of  $k$  capture more local features, such as noise or sharp transitions in the function value. The 0th component,  $X_0$ , is unique in that it represents the average value of the analyzed function, and remains constant over its domain. For discrete samples of functions, such as the  $N$ -dimensional vectors representing nodal force functions, we use the discrete Fourier transform (DFT), to extract these coefficients with respect to a given input function,  $f$ :

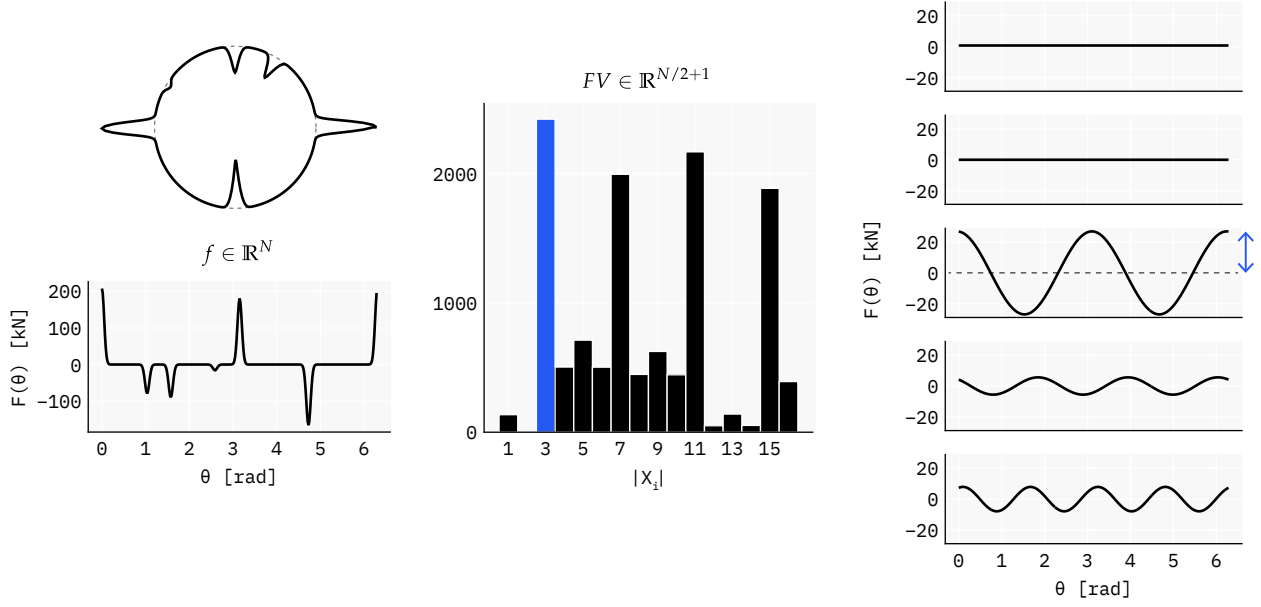


FIGURE 4.6: A nodal force function, the magnitudes of its Fourier transform coefficients, and the corresponding weighed cosine functions. A larger magnitude,  $|X_i|$  corresponds to a larger contribution of a given frequency cosine in the decomposition of the original function.

$$X_k = A_k + ia_k = \sum_{j=0}^{\frac{N}{2}} f e^{-i2\pi k \frac{j}{N}} \quad (4.4)$$

Unlike in general Fourier analysis, which operates on continuous, infinite-dimensional functions, the DFT performs an approximation of the transform over a finite number of samples,  $N$ . We further take advantage of the fact that all nodal force functions are real-valued, and leverage the symmetry of the cosine functions to reduce the overall number of summations by half [139]. Given a discrete vector representation of a nodal force function,  $f \in \mathbb{R}^N$ , the DFT returns a new vector,  $\mathbf{X} \in \mathbb{C}^{N/2+1}$ :

$$\mathbf{X} = [X_0, X_1, \dots, X_{N/2}]$$

The critical observation is that since Fourier analysis is a decomposition of an input function in the domain of *frequencies* rather than time or angles, the magnitude of the coefficients,  $|X_k|$  are invariant with respect to the initial orientation of the input function. In other words: rigid shifts of the input function, corresponding to rotations of the reference frame of a given nodal force demand, do not affect the output of the Fourier transform. This gives the final Feature Vector,  $FV$ , representation of the demand of a planar truss connection, which provides the necessary conditions of dimensionality and invariance for comparison:

$$FV = [|X_0|, |X_1|, \dots, |X_{N/2}|] \quad (4.5)$$

In Fig. 4.6, we show an example of how this decomposition is performed. The unwrapped force function is shown on the left, with the first 16 components of the Fourier coefficients



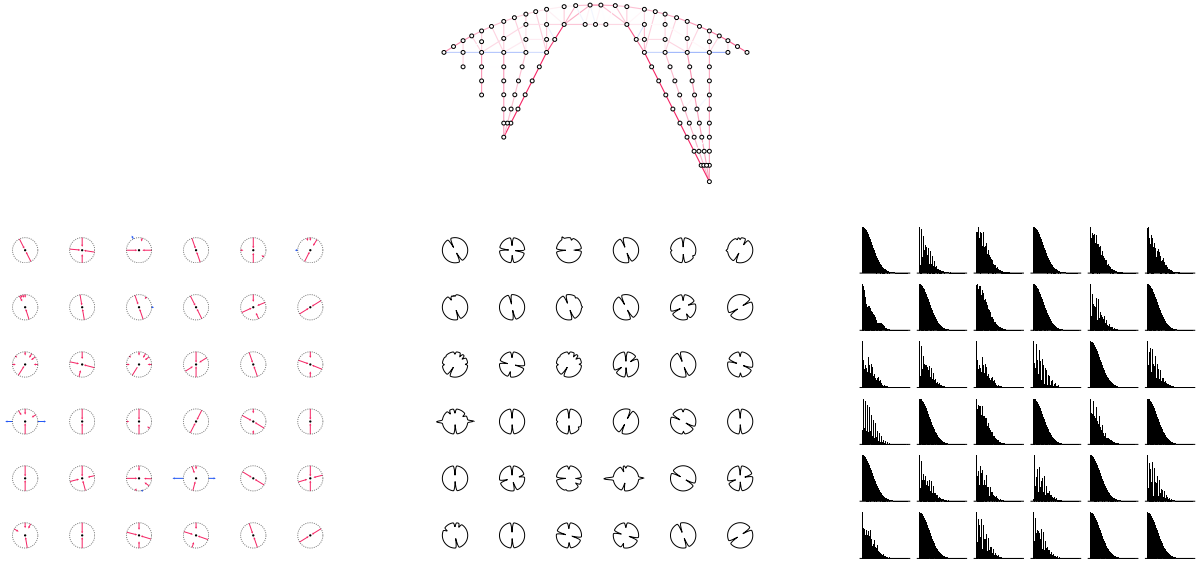


FIGURE 4.7: Three equivalent representations of the structural demand acting at a truss node: (1) explicit vector representation of forces, (2) functional representations enabling a consistent dimensionality independent of valence, and (3) magnitudes of Fourier coefficients, enabling a rotation-invariant representation of the force functions.

shown in the middle. On the right, we show the first five of the corresponding cosine functions that make up the original force function; larger values in the feature vector correspond to a larger contribution of the corresponding basis cosine function. In Fig. 4.7, we show the resulting feature vectors of a set of nodal force demands in relation to previous representations. The explicit vector representation on the left provides the most direct information about the magnitudes of forces acting on a node, as well as their orientations, but does not provide an efficient means of systematically comparing them. The function representation provides a similar level of information, but enables a more consistent numeric representation for computational analysis.

Although the dimensions of each force function representation can be held uniform by taking a constant sample rate for each node, they cannot yet be directly measured against each other, as the orientation of the forces acting on a given node will directly affect the order of values in each representative vector. Through Fourier analysis, the feature vectors provide the final level of abstraction needed to directly measure notions of similarity and uniqueness for truss connections. Although more visually abstract compared to the first two methods, it provides a compact and consistent means of capturing all relevant information about nodal force demand, and can be used for direct comparison. As the values in these feature vectors are only dependent on the magnitudes and *relative* orientations of the forces acting at a node, the dissimilarity between nodes can be directly computed through the Euclidean distance ( $L^2$  norm) of their corresponding feature vectors:

$$\text{dissimilarity} = \|FV_1 - FV_2\| \quad (4.6)$$



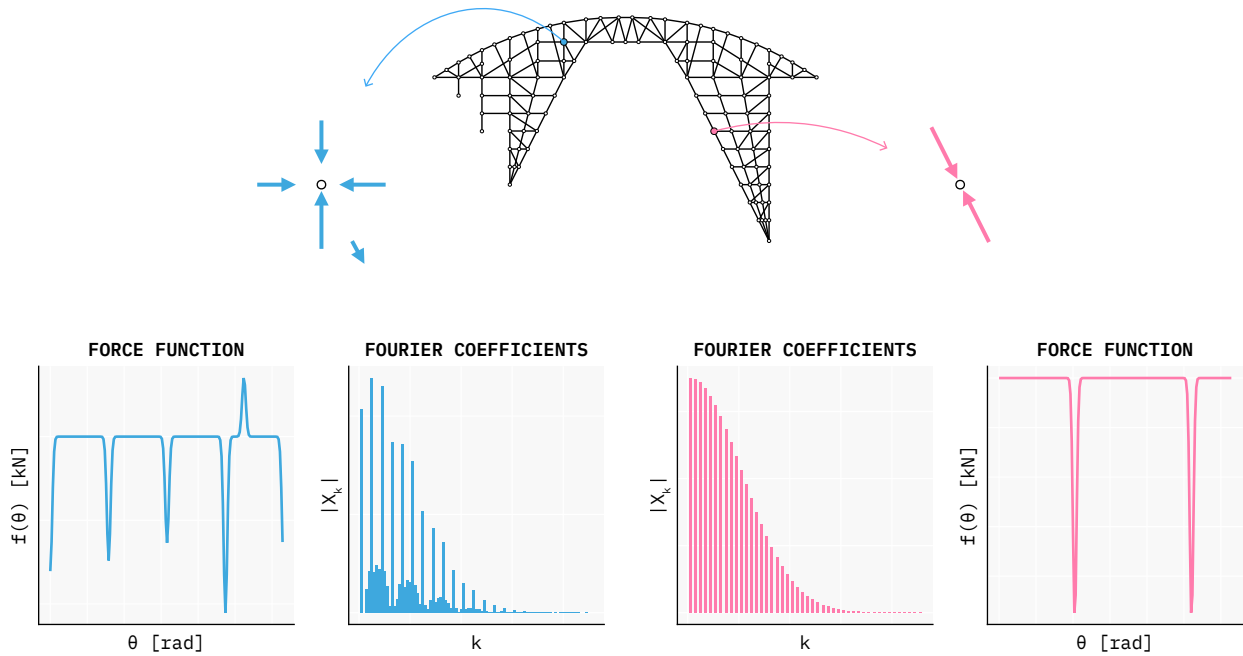


FIGURE 4.8: An example of two nodes, their corresponding nodal force demands, force functions, and their Fourier coefficients. The difference between the nodes can be measured by the Euclidean norm between these vectors.

In Fig. 4.8, we show this comparison explicitly through the comparison of two arbitrary nodes in our example structure, one with a valence of five and the other with a valence of two. We emphasize again that this difference in valence was the primary barrier to a simple comparison, as both nodes cannot be effectively represented in a common dimensional space. Their corresponding force functions alleviate this issue to a degree; given a common same size,  $N$ , their discrete functions could be compared through a vector norm. However, the orientations of the acting forces directly affect the values in the resulting vectors, akin to circular shifts along the independent axis of the force function, which may result in vastly different notions of difference depending on the initial alignment of the two nodes. The Fourier coefficients, or the feature vectors, provide a more abstract but directly measurable vector representation of the underlying force functions, enabling a direct distance comparison as a measure of similarity.

#### 4.5 NODAL DEMAND COMPLEXITY: MEASUREMENT, VISUALIZATION, AND RATIONALIZATION

One challenge with the  $N$ -dimensional feature vector representation of nodal force demands is that they are difficult to directly interpret at the scale of an entire structure. In early stage design, understanding the influence of structural form on the distribution of nodal force demands enables a more informed design process that can be tailored to the constraints of a given project. Although we cannot directly show all  $N$  dimensions of the feature vectors,

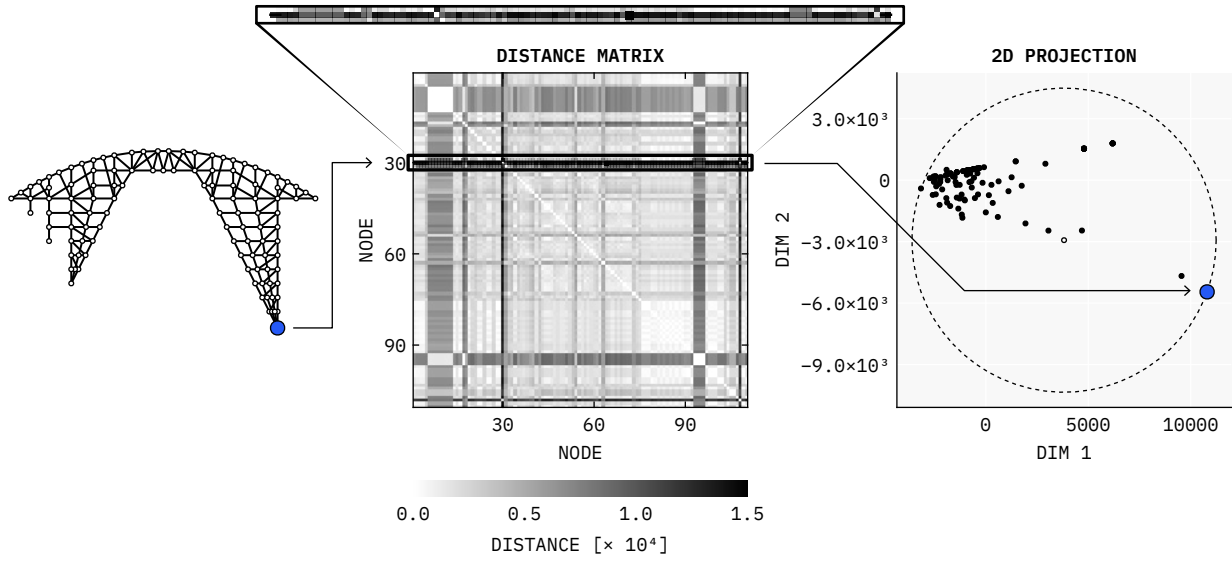


FIGURE 4.9: Visual representations of nodal force demand variation through: (1) a distance matrix that captures all inter-node dissimilarities, and (2) a projection of the original feature vectors into a 2D space through Multidimensional Scaling (MDS).

two visual cues can be still be provided, shown in Fig. 4.9. First is the distance matrix, DM, shown in the center of the figure, which shows the magnitude of Eq. (4.6) between each pair of nodes in the structure, where:

$$DM_{ij} = \|FV_i - FV_j\| \quad (4.7)$$

Each row or column corresponds to a given node in the structure, and the color represents its relative dissimilarity to the remaining nodes in the structure. Although difficult to identify which node in space corresponds to a given row, the distance matrix provides a broader overview of the overall distance landscape, and enables the identification of outliers. For example, the dark band in row 30, corresponding to the bottom-right support in the structure, indicates a highly unique demand condition, with little similarity to any other node in the structure.

The second visual interpretation is through a projection of the original high-dimensional feature vectors into a more interpretable lower-dimensional space, such as  $\mathbb{R}^2$ . Using Multidimensional Scaling (MDS) [140], we can transform the computed distance matrix into a set of points in  $\mathbb{R}^2$  that best preserves the original inter-point distances; this visualization is shown on the right of Fig. 4.9. Consistent with the darker band in row 30 of the distance matrix, we can observe this same uniqueness of this nodal force demand by its equivalent point in the 2D project, separated from the main cluster of points, and only similar to the other ground support node. A scalar measure of general nodal demand variation can also be measured, both in the original high-dimensional space and in the projected 2D space using MDS, by measuring the minimal bounding  $n$ -sphere of the set of feature vectors. Shown as the dashed circle in the projected space, the size of this circle provides a measure of the spread of nodal force demand conditions as an indicator of manufacturing and

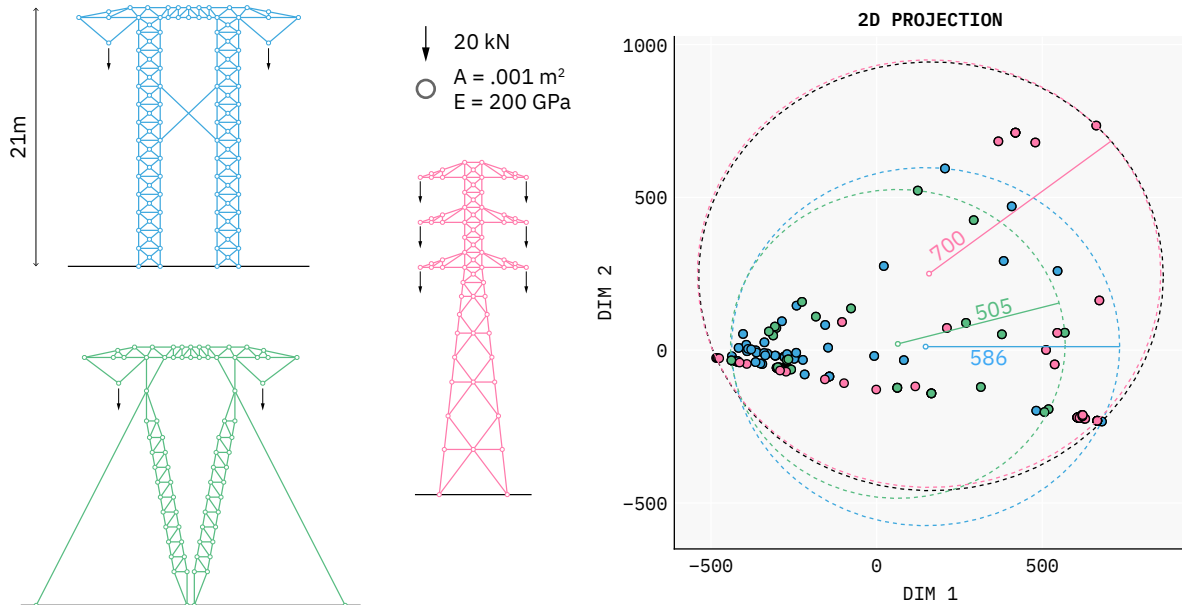


FIGURE 4.10: Three competing transmission tower designs and their resulting nodal demand distributions (projected in 2D). The complexity score for each design is shown by their respective circles. These metrics can be used for design evaluation and comparison when connections are important in the design process. The black dashed circle represents the bounding circle of all three designs combined.

assembly complexity. Although the magnitude of this value, as well as the numerical values of the feature vectors, do not have a direct physical significance, their *relative* values when comparing designs or pairs of nodes provides a useful measure of relative similarity.

Although the feature vectors present a level of abstraction beyond physical geometry and orientation, through both the distance matrix and the MDS projection, they can nonetheless provide structural designers with direct insight throughout the design process on the effect of global form on the distribution of nodal force demand. When comparing a range of design options, this method can also provide a means of quick evaluation of nodal complexity, as shown in Fig. 4.10. Here, three different transmission tower designs are shown, along with their projected feature vectors and bounding circles. Should standard node detailing be a concern in the design process, the relative sizes of these circles can be used to observe the effect of each design on nodal demand complexity. The projected visualization can also provide a measure of general similarity between the distribution of nodal force demands across the three designs. If all three designs must be constructed using similar connection details, the complexity scores may be used to drive the designs towards similar distributions of node force demands, enabling the use of a consistent base of connections across all structures.

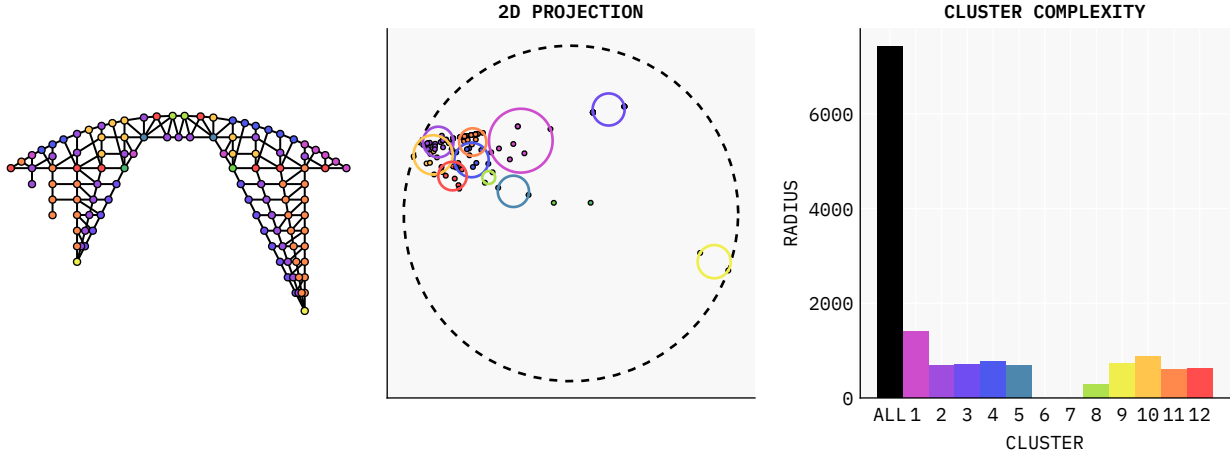


FIGURE 4.11: Nodes clustered by the similarity in the feature vector space, with associated complexity score.

#### 4.6 REDUCING COMPLEXITY

Given a measure of the variation of nodal force demands, a natural target is in its reduction. As before, one possible method of complexity reduction is through the clustering of nodes into groups of similar demand conditions. Nodes that experience similar element orientations and force magnitudes could be sized and detailed in conjunction, enabling reduced effort in individual detailing and fabrication. In Fig. 4.11, the initial distribution of nodal force demands are clustered into 12 individual groups that minimize group wise complexity using k-means clustering [74]. Here, the bounding sphere metric for complexity is used as it provides a clearer visual representation of cluster-wise complexity.

Both visually and numerically, the clustered node groups reduce the overall spread in the feature vector space, corresponding to a higher degree of similarity between nodes in each group. This reduction in complexity can be directly observed through the arrow representation of nodal demand in Fig. 4.12. Beginning with the fully superimposed set of nodal force demands in the top of the figure, we then show the collection of force vectors grouped by their cluster assignment. Generally, each cluster reduces the distribution of force angles acting at the clustered nodes, allowing for a clearer delineation between individual forces. For example, Cluster 9, which groups the two primary supports of the main arches, both consist of a highly-bundled group of elements that enter from the top of the node, with a single large reaction opposing their summed vectors.

The clustered force groups can directly be used for the design of connections, as shown in Fig. 4.13. Here, we leverage once again the force functions of each cluster, as they already provide a smoothed representation of the force demands through the Gaussian bumps. By simply summing these functions over the unit circle, we can generate a cumulative force function,  $f_c(\theta)$ , that represents all forces acting at a given cluster:

$$f_{ci}(\theta) = \sum_{i=1}^{n_c} f_i(\theta) \quad (4.8)$$

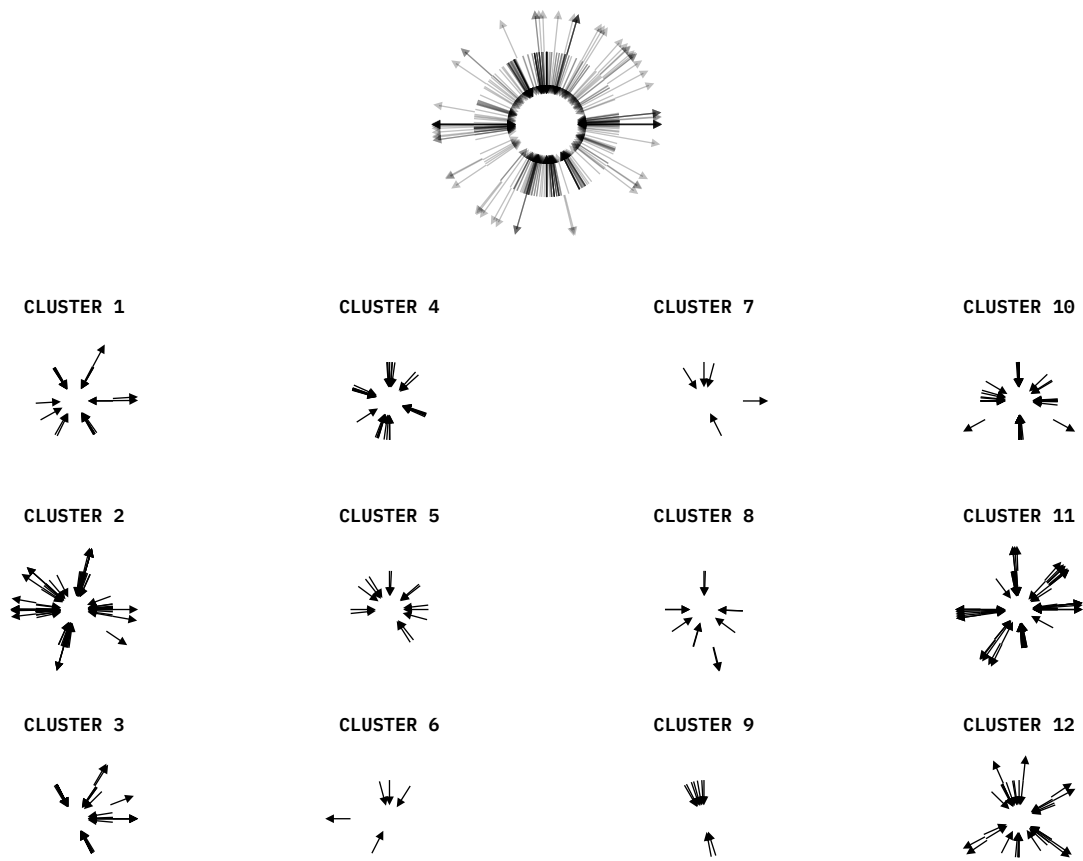


FIGURE 4.12: Clustered nodal force functions visualized as arrows, with the length of each arrow corresponding to the magnitude of the force at that angle.

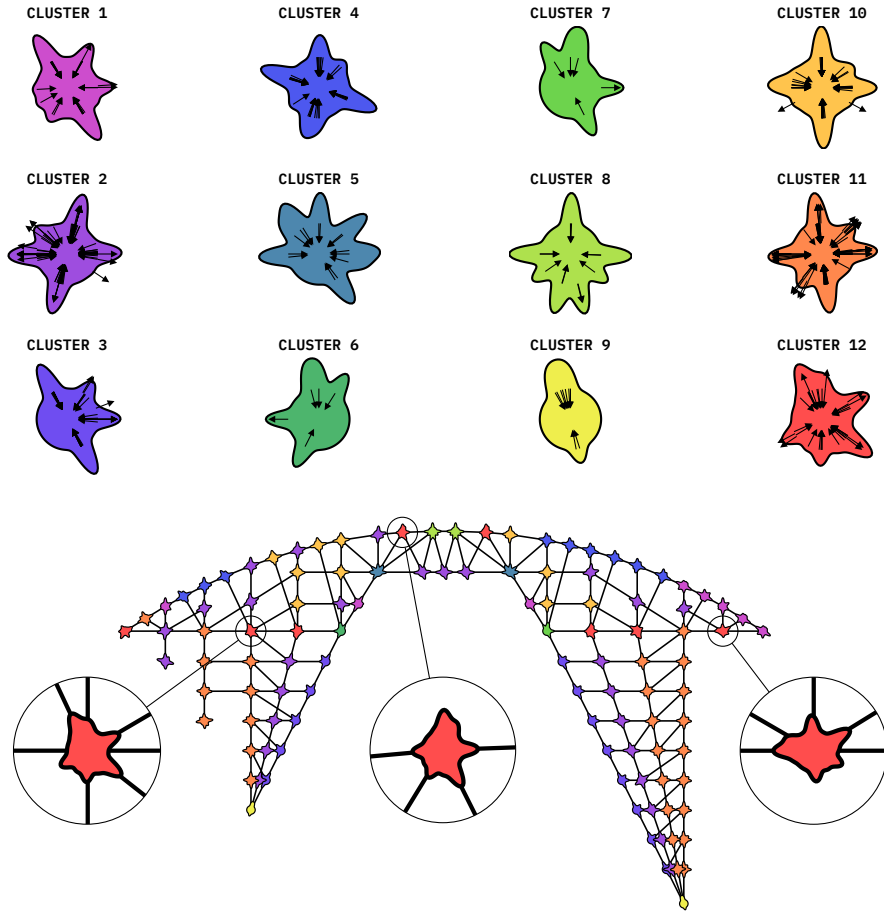


FIGURE 4.13: An example of 12 clustered nodal force demands converted into gusset plate geometries to construct the entire structure. We highlight how the design for Cluster 12 can be used in a variety of orientations.

By nature of the Gaussian distributions, forces that are close together in orientation will naturally generate an aggregated bump concentrated at their average angle, while maintaining the smoothness of the individual functions. This smoothing can also be directly tuned by adjusting the  $\delta$  parameter of the individual force functions, as described in Fig. 4.5; the magnitude of this parameter directly controls the degree to which closely-aligned forces are merged together. An example of this result is shown in Fig. 4.13. When wrapped around a circle, with  $|f_{ci}(\theta)|$  representing a radial offset from an initial circle radius at  $\theta$ , the cumulative force function provides the natural boundary for a shaped gusset plate, whose thickness can be calibrated by the magnitudes of the forces acting at each cluster. Here, we show how a structure of 110 nodes can be adequately constructed with 12 unique gusset plate geometries, which can be detailed and fabricated at scale to reduce the overall construction complexity. Highlighting Cluster 12 (red), we show how the aggregated gusset geometry can provide proper element registration and alignment for wildly varying node valences and element orientations.

Our choice of 12 clusters in this example is arbitrary. In practice, this value should be calibrated to the degree of complexity that is acceptable for a given project, with higher

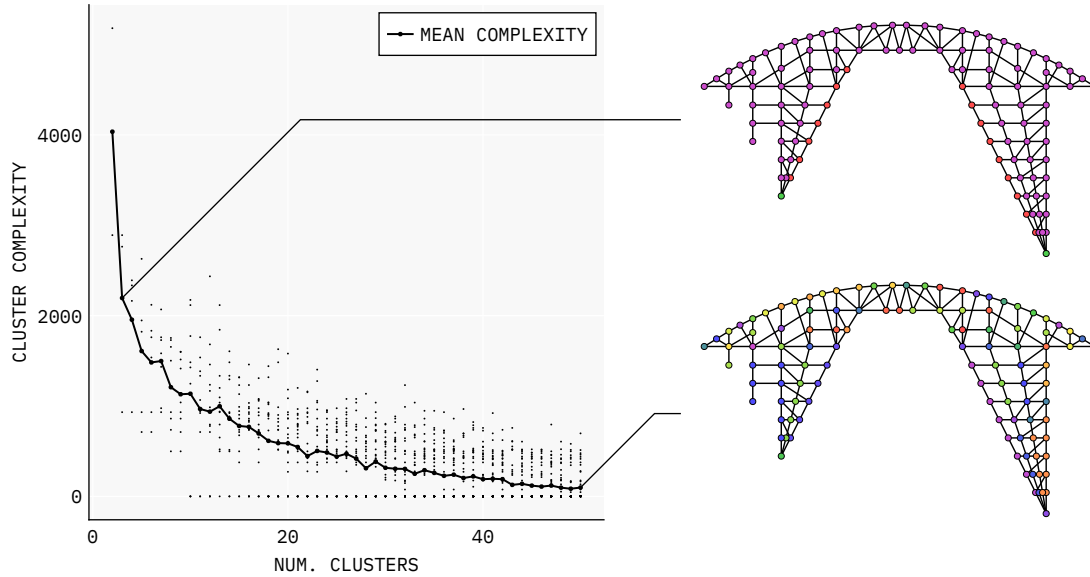


FIGURE 4.14: Relationship between the number of clusters and the mean complexity of the resulting clustered force demands. The individual complexity values of each cluster are also shown as discrete points.

values indicating a larger bandwidth of effort in the design and detailing of individual connections. However, there will be a natural trade-off between this value and the overall reduction in complexity. In Fig. 4.14, we perform the same exercise for a range of cluster counts, ranging from 2 to 50. Individual cluster complexity values are shown as discrete scattered points, with the mean complexity value plotted as a scattered line. Although it is expected that mean complexity would decrease with more clusters, this relationship is not linear, and provides a diminishing returns at large values. As for many multi-objective design problems, a clear starting point for clustered node design may be at a visual knee point, which can then be calibrated by the designer with the input of fabricators and other stakeholders. Our method enables this evaluation to occur in an automated and intuitive manner, while maintaining higher-level decision-making in the hands of the designer.

#### 4.7 STRUCTURAL PERFORMANCE AND NODAL COMPLEXITY

Of course, nodal complexity is not the only consideration in the design and construction of both efficient and feasible truss structures. The dominant mass, and resulting embodied carbon, remains in the structural elements themselves, suggesting that their minimization remains the priority. In this section, we explore the relationship between nodal complexity, global material efficiency, and the complexity of the resulting Demand Space of truss elements. In Fig. 4.15, we show a doubly-cantilevered Warren truss structure under load, first shown in Chapter 2, and its subsequent shape optimization problem formulation.



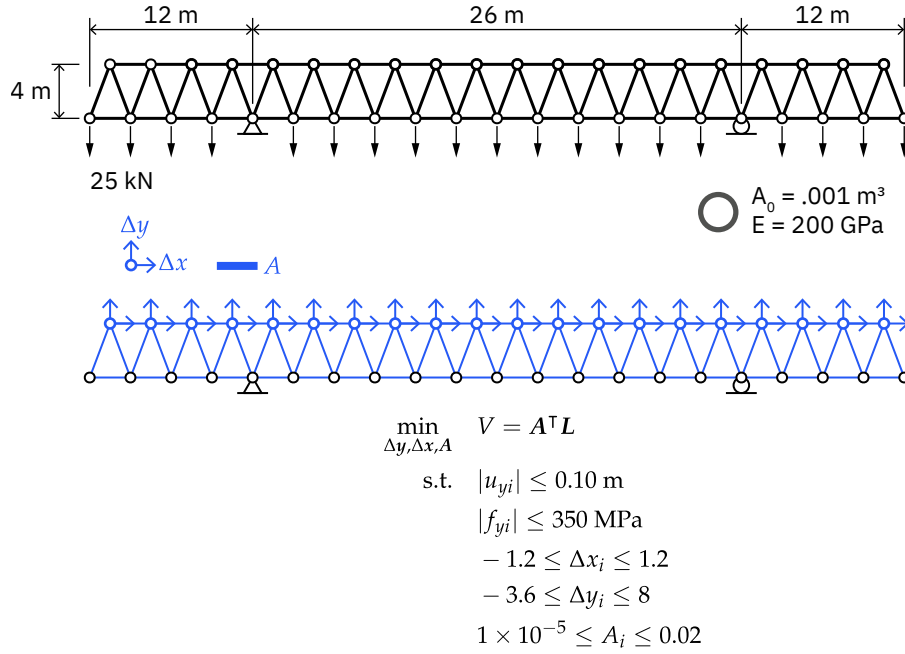


FIGURE 4.15: A Truss structure and its minimum volume optimization problem formulation.

We seek a minimum volume design through the nodal positions of the top chord nodes, as well as the cross-sectional areas of all elements, while maintaining typical stress and displacement constraints across the structure. The explicit formulation of the problem is:

$$\begin{aligned}
 &\min_{\Delta y, \Delta x, A} V = A^T L & (4.9) \\
 &\text{s.t. } |u_{yi}| \leq 0.10 \text{ m} \\
 &\quad |f_{yi}| \leq 350 \text{ MPa} \\
 &\quad -1.2 \leq \Delta x_i \leq 1.2 \\
 &\quad -3.6 \leq \Delta y_i \leq 8 \\
 &\quad 1 \times 10^{-5} \leq A_i \leq 0.02
 \end{aligned}$$

Using the Method of Moving Asymptotes (MMA) algorithm [77] and differentiable structural analysis, we solve this problem in 309 iterations and 22 seconds; the optimization trace and intermittent designs are shown in Fig. 4.16.

The optimal solution provides an expected chord geometry that reflects the moment distribution across the structure, with the deepest separation between top and bottom chords occurring both at the supports and at mid-span. The optimal cross-sectional areas, whose relative differences are shown by the line weights of the drawn elements, also show larger areas assigned to the chord elements, with smaller areas assigned to the web members. Ultimately, this solution results in an 86 % reduction of the initial volume of the structure, from 0.28 m<sup>3</sup> to 0.04 m<sup>3</sup>, with a corresponding reduction in embodied carbon. A majority of this improvement occurs early in the optimization trajectory, as highlighted by the steep

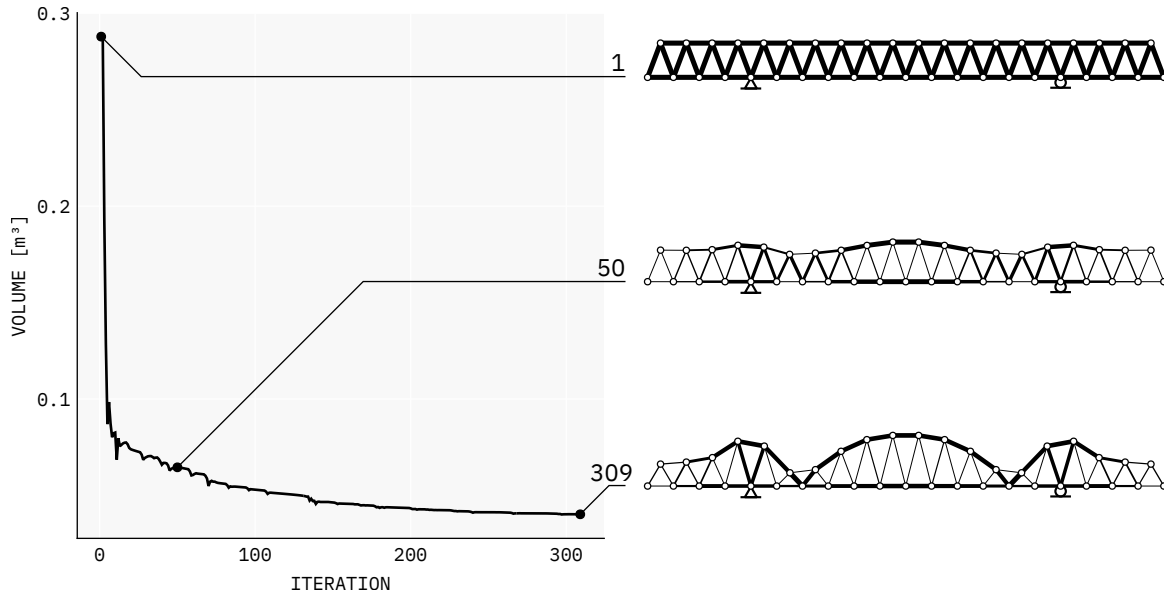


FIGURE 4.16: Optimization trace of the initial design problem.

initial decline of the objective trace, and the structural form and material distribution shown at iteration 50, shown in the middle of Fig. 4.16. The remaining iterations focus on local refinements of geometry, primarily the tapering of the two cantilevered ends of the chord to reflect the low moment demand.

However, to achieve this 86 % reduction in theoretical volume comes at an expected cost of complexity. By shaping the truss to minimize internal forces, the angles between elements at the connections become more varied, especially compared to a highly regular Warren truss geometry. To evaluate this impact, specifically of the geometry of elements, the feature vectors of the *normalized* nodal force functions are taken at each iteration during optimization. Normalization set the magnitude of all peaks and values in the force function to 1, ensuring that each incoming element has the same influence on the overall demand function, regardless of that element's magnitude; this in turn enables the computation of a purely geometric feature vector representation of each node, from which the complexity can be computed again as the radius of the bounding sphere of the set of vectors. In Fig. 4.17, the projected feature vectors of nodal force demands are shown for both the initial and final design states, alongside the corresponding elemental feature vectors. The normalized measures of the structural volume, as well as the nodal and elemental demand complexities, are shown at the top of this figure.

There is a clear, but not smooth, inverse relationship between global structural volume and the complexity of the structure, both in the elements and their connections. As the geometry of the truss transitions from a highly-structured form to one that better transfers the flow of forces towards supports, varying inter-element angles cause the four unique node conditions (support, bottom chord, top chord, and two corners) to expand into a larger variation of demand. This is reflected by the increase in the number of unique feature vectors in the projected Demand Space of nodal demand, and a corresponding increase in the size of the bounding sphere. The increase converges to a consistent value in the latter half of optimization, and the peak measure of complexity occurs in the earlier stages

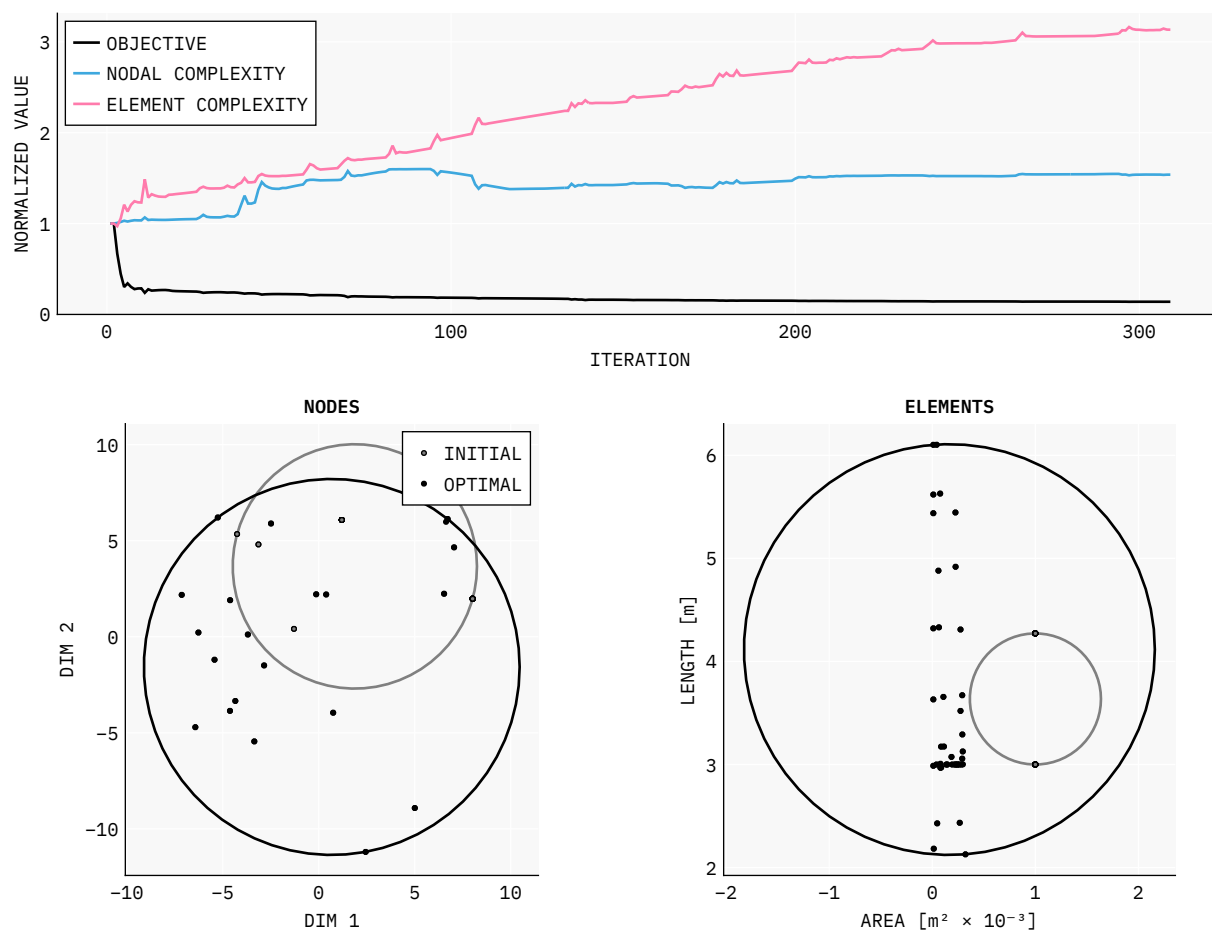


FIGURE 4.17: *Top*: Normalized objective function value, nodal feature vector complexity, and elemental feature vector complexity throughout the optimization trajectory. *Bottom*: The feature vectors of the nodal force demands and elemental area and length feature vectors at the initial and final design stages.

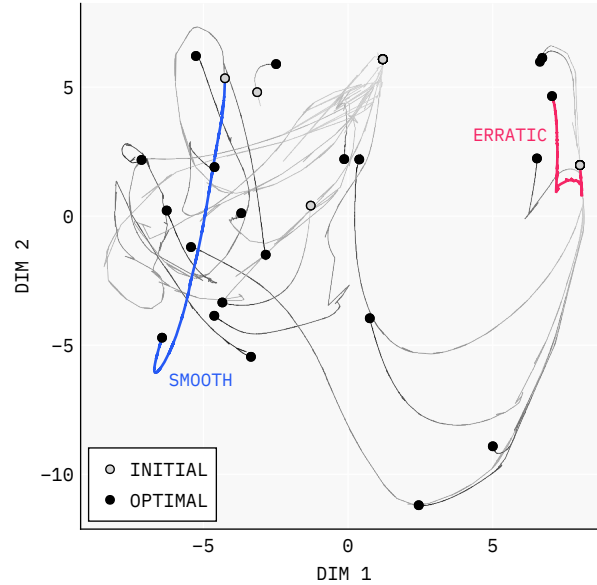


FIGURE 4.18: The projected nodal force demand feature vectors throughout the optimization trajectory.

of optimization. Further, the first few iterations of optimization, which shows the largest reduction in the structural volume objective function, shows very little change in modal force demand, as this region is driven by changes in element area rather than the geometry of the truss.

As discussed in Chapter 2, the bounding sphere radius is a metric of complexity that is focused on extrema in the Demand Space. One may wish to observe how individual feature vectors change throughout the optimization trajectory, and observe local changes and trends. In Fig. 4.18, we plot the trajectory of each feature vector from the initial distribution (in gray) to the final distribution shown in black. Although the overall structure itself exhibits continuous, smooth changes in its geometry and section areas due to the gradient-based nature of optimization, the trajectory of nodal force demands are more erratic. Some nodes show smooth arcs in the Demand Space, reflective of a smooth change in similarity from other connections; other nodes exhibit sharper transitions, and sometimes revert to previous states passed during optimization. Two examples are highlighted in Fig. 4.18.

In this example, we have observed an inverse relationship between global performance and component-level complexity. Minimum-volume structures, which seek to minimize stress distributions by generating smooth pathways from loads to supports, will naturally result in more complex element lengths and inter-element orientations at the connections. Optimality is typically achieved at an aggregated scale, where the shaped geometry concentrates material at key locations while minimizing stress distributions elsewhere. This comes at the cost of a larger variation in demand, which affects the design's ease of construction. However, by capturing this relationship between design and demand, the designer can make informed decisions about these trade-offs early in the design process. One could evaluate the optimization history and identify designs where minimal complexity is added while still achieving significant reductions in structural volume. Alternatively, one could

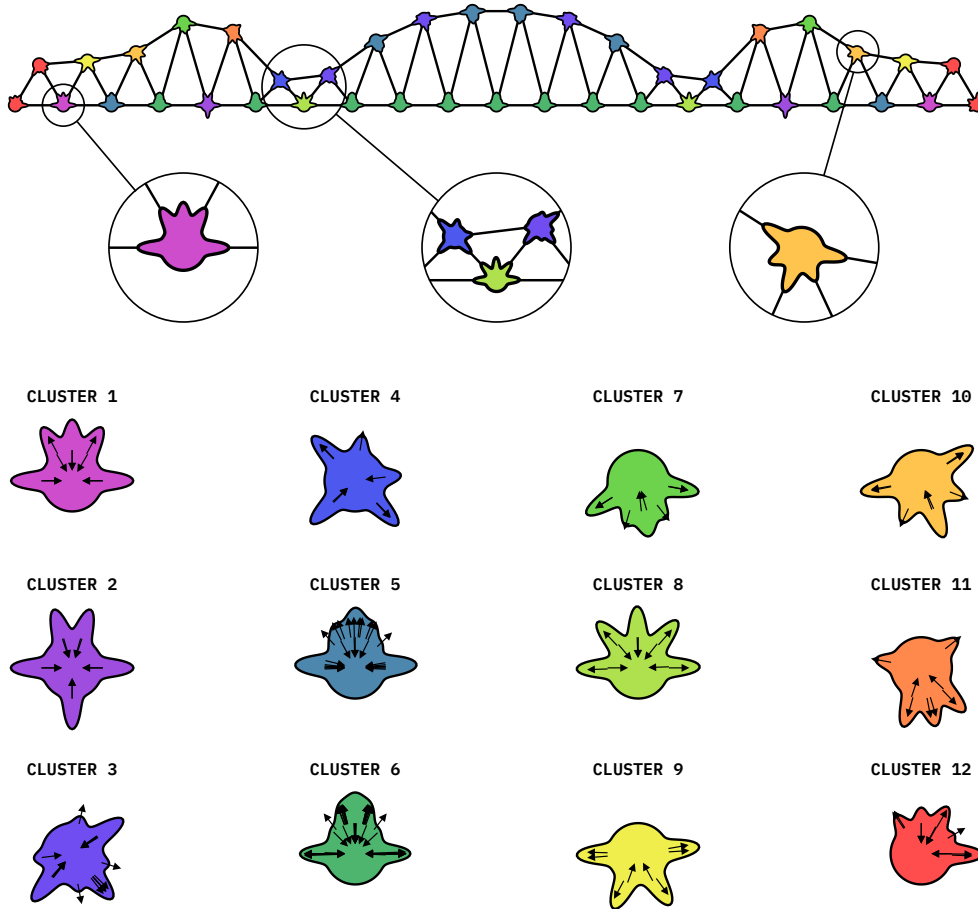


FIGURE 4.19: The optimized truss with clustered nodal force demands and corresponding gusset plate geometries. *Note:* the bottom chord nodes contain the external loads (applied from the top) as well as support reactions (applied from the bottom).

rationalize the outcome of the optimized result by identifying clusters of similar nodal force demands, and reducing the number of unique connections required for sizing. In Fig. 4.19, we show again how clustering can be used to detail a reduced set of gusset plate geometries for the optimized truss structure.

#### 4.8 CONCLUSION

Connections, despite taking a small proportion of overall structural volume, nonetheless play an important role in a structure's function and ease of construction. As physical components that must be designed, detailed, and fabricated, they too should be a consideration in early in the design process when considering the effect of global form on design complexity. However, unlike structural elements, which contain a finite set of relevant information that make up its demand, defining a consistent representation of nodal force demand is more challenging, as they must account for varying valences, orientations, and magnitudes of forces acting at a given connection. This chapter demonstrates how harmonic analysis can

be leveraged to overcome these challenges, by interpreting nodal force demands as smooth, periodic functions on the unit circle. By taking the Fourier transform of this functional representation, we can represent nodal force demand as a compact  $n$ -dimensional feature vector that is invariant to node orientation and valence. The collection of these feature vectors form the Demand Space of planar truss connections, from which distance and similarity can be consistently measured and compared.

One caveat in this method is the inherent symmetry of the Fourier transform based feature vectors used to generate the Demand Space of nodal force demands. As each component of the feature vector is computed from the *magnitude* of the resulting complex Fourier coefficients, there is a loss of information in the direction of the phase shift of each representative frequency, represented by the imaginary component of the coefficient. This means that nodes that have a reflected demand condition will result in the same feature vector representation, even if there is no possible rigid rotation of one force function that can be applied to perfectly align one node with the other. This is best exemplified by the resulting gusset plate design in Cluster 10 in Fig. 4.19, where there exists a superfluous limb that can be removed by simply reflecting the force function across the planar axis, akin to flipping the physical gusset plate to its other side. In practice, this flipping procedure could be performed as a post-processing step after clustering to ensure the simplest connection geometry that can be used for all nodes in a given cluster; however, for clusters with a significant number of node conditions, this amounts to a combinatorial search exercise, which may incur computational and time costs.

Beyond this limitation, the method presented in this chapter provides a method of understanding and operating on the Demand Space of connections to visualize, measure, and mitigate design complexity. A shift in perspective from discrete sets of nodes and angles to a continuous function representation of nodal demand enables the use of harmonic analysis to generate a Demand Space robust to topology, valence, and orientation.